

Overview of the Global Drone Industry

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Abstract—This review introduces the trends of the global drone industry and the manufactures, the forecast of drone industry, the use case, the ranking of global drone performances including AAM (Advanced Air Mobility).

Keywords—UAV, Drone, trend, market, manufacturers, forecast, ranking of drone performance, AAM (Advanced Air Mobility)

I. INTRODUCTION

This review paper provides a bird's-eye view of the latest trends in the global drone industry. In particular, it provides a detailed analysis of the global drone industry overview, trends among drone manufacturers[1], and the current and future state of global Advanced Air Mobility(AAM) which is sometimes called as "flying car" technology. Specifically, the current state of the global drone industry is explained with the database of Drone Industry Insight, a German company. Next, a database based on original research is constructed to analyze and consider the trends of global drone manufacturers, including flight performance with ranking. Furthermore, the current state and future state of global AAM technology is explained from the perspective of flight performance, although all of them are still under development, as this is the field that is evolving the most and attracting the most attention.

II. CURRENT STATE OF THE GLOBAL DRONE INDUSTRY

According to a document released on September 14, 2022 by Drone Industry Insights (DII)[2], a company based in Frankfurt, Germany, the market size of the global drone industry is \$30.6 billion in 2022, as shown in Fig. 1. In 2030, it is expected to reach \$55.8 billion with an annual growth rate of 7.8%. Meanwhile, a report from the same DII in 2019 reported that growth of more than \$43.1 billion is expected in 2024. The company also predicted a compound annual growth rate (CAGR) of 20.5%. Despite the global spread of the COVID-19 pandemic, DII has significantly revised its market forecast downward from a CAGR of 20.5% in 2019 to 7.8% in 2022. This means that the global drone industry has moved from a period of rapid growth to a period of stable growth.

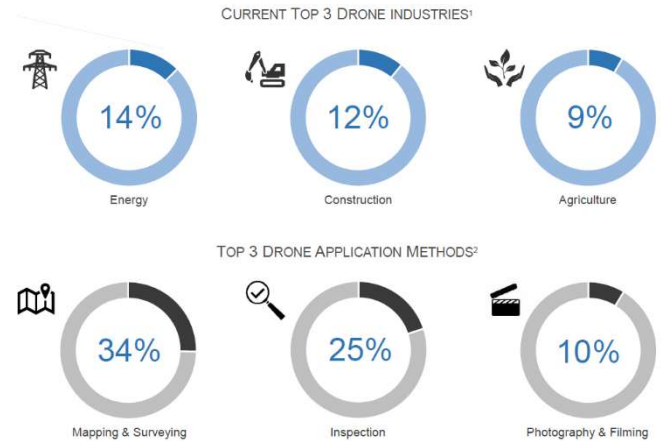
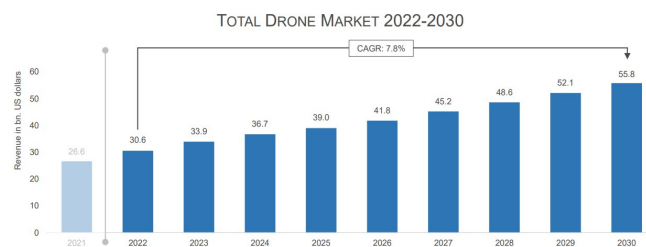


Fig. 1 Global drone industry market growth forecast[2]

Fig. 2 Top 3 drone industry and application methods[2]

Fig. 2 shows the top three drone industries and the top three drone applications. The energy sector is in first place, followed by the construction sector and the agriculture sector. In terms of drone usage, mapping and surveying, facility inspection, and aerial photography are the top three.

Fig. 3 shows the number of drone job openings by occupation in each country and the number of job openings by occupation in the world. From this figure, the United States stands out in the number of job openings, and when we look at the details, we can see that the most job openings are in drone logistics services, software, AAM, and drone hardware, in that order. In the case of Germany, which ranks second in the number of job openings, the number of job openings related to AAM is abnormally high in AAM and drone hardware. In third place, India has drone hardware, in fourth place, Australia has logistics drones, and in fifth place, Switzerland has hardware and software, which reflects the actual situation of each country. In China, drone hardware and AAM are in a close competition, which well reflects China's manufacturing industry. In Malaysia, drone services are overwhelmingly more common. Japan is ranked 10th in the world in terms of the number of job openings, and is characterized by drone services being the main job field, so the image of "Japan, the manufacturing industry" is not necessarily visible. In terms of the number of job openings by occupation in the world, the order is software and marketing, which shows a tendency to differentiate drones with drone software.

Fig. 4 shows the reasons for adopting drones. In order of importance, the most common reasons cited were improving safety, saving time, improving quality, and reducing costs.

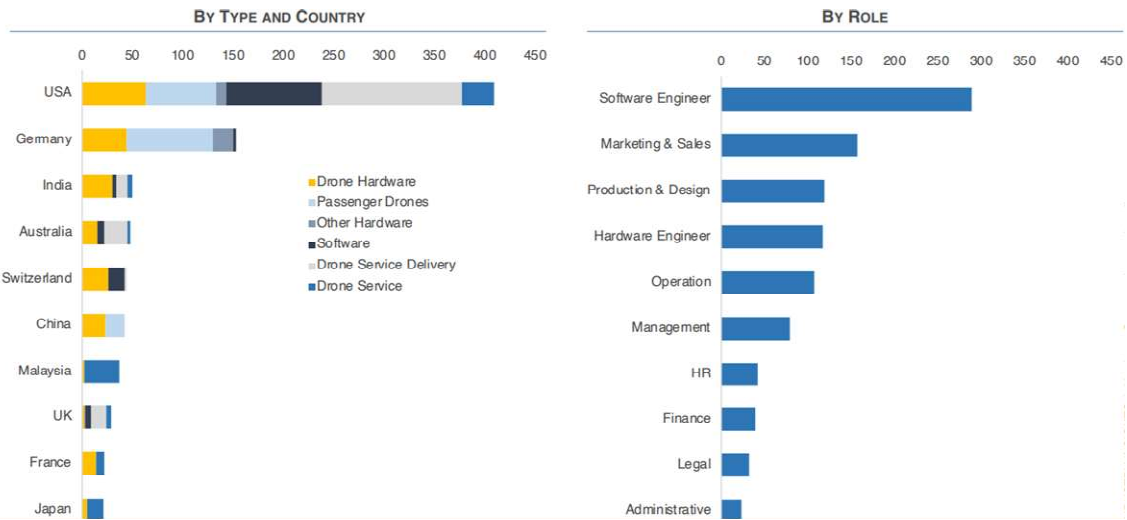


Fig. 3 Number of drone job openings by occupation in each country and by occupation worldwide[2]

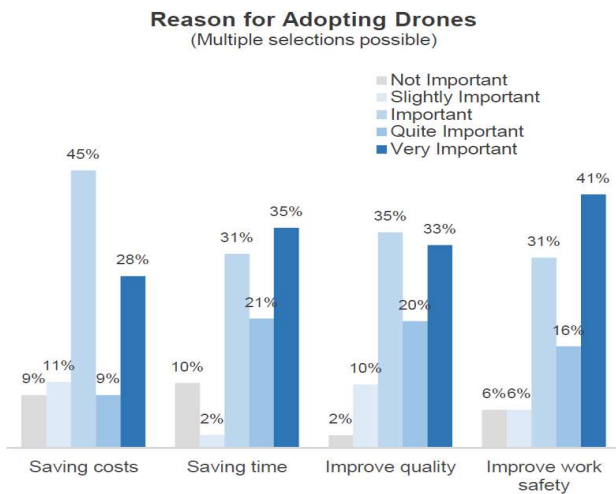
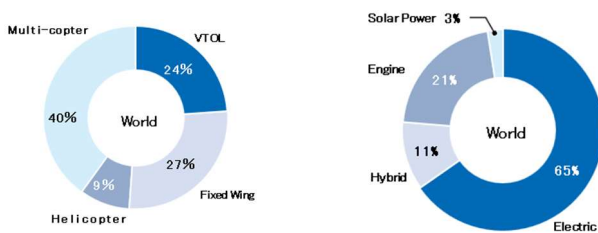


Fig. 4 Reasons for adopting drones[2]

III. TRENDS OF GLOBAL DRONE MANUFACTURERS

The mainstream of drones in the world are fixed-wing + VTOL, and the breakdown by type of aircraft of interest is shown in Fig. 5(a) (AAM are excluded). Multi-copters are the most common at 40%, followed by fixed-wing at 27% and VTOL (Vertical Take-Off and Landing) at 24%. If we divide fixed-wing + VTOL types, which obtain lift during cruising from fixed wings, as "fixed-wing types," and helicopter + multi-copters, which obtain lift from rotors, as "rotor types," then fixed-wing types account for 51%.

The breakdown by power source is shown in Fig. 5(b) with 65% of aircraft being electric, the largest share,



(a)Percentage of drone type (b)Percentage of power source
Fig. 5 Percentage of drone type and power source

followed by engines at 21% and engine + electric hybrids at 11%. If the 3% solar power generation is added to the electric version, the figure is 68%, and if hybrids are also considered "electric," then 79% are electric.

Table 1 shows the ratio of power sources by type. Here, fixed-wing types are divided into large and small, with a body weight of 10 kg as the dividing line, because the operation methods are very different between large aircraft that require runway takeoff and small aircraft that can be hand-launched (hand-thrown drones) or bungee-launched (tethered drones). While fixed-wing (small) types and multi-copter types are almost all electric, Engine drive is the mainstream for fixed-wing (large) and helicopter types. For VTOL types, nearly 70% are electric, but the proportion of hybrids is also high. Hybrid systems include aircraft that are motor-driven only during takeoff and landing, and engine-driven during horizontal flight cruising speed, or aircraft that generate electricity while the engine is running and store it in a battery.

TABLE 1 Power source ratio by drone type

	Electric	Hybrid	Engine	Solar Power
VTOL	67 %	23 %	10 %	0
Fixed Wing(large)	8 %	14 %	70 %	8 %
Fixed Wing(small)	95 %	0	0	5 %
Helicopter	33 %	0	7 %	0
Multi-copter	90 %	9 %	0	1 %

IV. GLOBAL DRONE FLIGHT PERFORMANCE

When it comes to flight performance, three indicators are important: maximum flight speed, longest flight time, and maximum payload. Here, we show the world rankings.

TABLE 2 Top 10 maximum speed flight performance drone

Type	Power Source	Model Name (Manufacturer, Country)	Maximum Speed
Helicopter	Engine	CAMCOPTER® S-100 (Schiebel, Austria)	240km/h
Fixed Wing	Hybrid	TALISMAN HD® (Elytron Aeronautica, Italy)	230km/h
Fixed Wing	Engine	The NASC Tero™ (Navier Applied Science, United States)	222km/h
Fixed Wing	Engine	BLACK SWAN (DRONAMICS, Bulgaria)	200km/h
VTOL	Hybrid	Kite™ (SwoopAero, Australia)	200km/h
Fixed Wing	Engine	KUS-FT (KOREAN AIR AEROSPACE, Korea)	200km/h
Fixed Wing	Engine	YZ8 (BMP, China)	190km/h
Fixed Wing	Engine	Atobá (Stella Tecnologia, Brazil)	190km/h
VTOL	Engine	FLEXROTOR (Aerovel corporation, United States)	180km/h
VTOL	Hybrid	Aero2 (Dufour Aerospace, Switzerland)	170km/h

Table 2 shows the top 10 in the world for maximum speed. Most of them are from European and American manufacturers. The aircraft are made by manufacturers in Korea and China, but there are also aircraft made by Korean and Chinese manufacturers. The top three are Austria, Italy, and the United States, with Bulgaria, Australia, and Korea in fourth place. In first place is the Camcopter from Austria, a single rotor helicopter driven by an engine. This is surprising, and overturns the preconception that fixed wings fly faster than rotary wings. Schiebel of Austria has been researching and developing rotary engine-driven rotors for many years, and it is believed that this result has led to this result. The second and subsequent places are fixed wing and VTOL aircraft, most of which are engine-powered or hybrid-powered. The speed is 240 to 170 km per hour.

TABLE 3 Top 10 maximum flight time performance drone

Type	Power Source	Model Name (Manufacturer, Country)	Flight Time
VTOL	Engine	FLEXROTOR (Aerovel corporation, United States)	33 hours
Fixed Wing	Engine	Atobá (Stella Tecnologia, Brazil)	28 hours
Fixed Wing	Engine	Raybird 3 (Skyeton, Ukraine)	18~28 hours
Fixed Wing	Engine	ORYX (CAT UAV, Spain)	26 hours
Fixed Wing	Engine	The NASC Teros (Navmar Applied Science, United States)	24 hours
Fixed Wing	Engine	H24CRUZER (Harpia Tech, Portugal)	24 hours
Fixed Wing	Engine	SkyRobot* FX450 (Robot Aviation, Norway)	20 hours or more
Fixed Wing	Engine	Penguin C MIL UAS (Edge Autonomy, United States)	20 hours or more
Fixed Wing	Engine	Patroller* (Safran, France)	20 hours
Fixed Wing	Engine	TEKEVER AR5 (TEKEVER, Portugal)	20 hours

Table 3 shows the top 10 aircraft with the longest flight time. All of them are engine-powered aircraft. The first place is a VTOL aircraft, but all the others are fixed-wing aircraft. The flight times are 33 hours, 28 hours, etc., and they can fly for more than a day. As with flight speed, most of the time, Europe and the United States dominate, but Brazil is also doing well in second place. The United States is always the favorite, but we can see that Ukraine, Spain, Portugal, and Norway are also doing well.

Fig. 6(a) shows the aircraft, performance specifications, and three main features of the Camcopter from Austria, which came in first. A single-rotor aircraft equipped with a 50hp rotary engine can be equipped with multiple sensors for security, infrastructure monitoring, surveying, investigation, etc. Highly weather-resistant with an operating temperature range of -40 to +55° C, and has a track record of operation in the Arctic Ocean. In particular, the first place aircraft, the American-made "FLEXROTOR", as shown in Fig. 6(b), is a tail-sitter aircraft, which takes off and lands vertically and can also hover in this position, making it a very unique fixed-wing aircraft.



(a) Camcopter (b) FLEXROTOR

Fig. 6 World top class drones with maximum speed or maximum flight time

has a flight time of over 12 hours.

TABLE 4 Top 10 maximum payload drone

Type	Power Source	Model Name (Manufacturer, Country)	Maximum Payload
Fixed Wing	Engine	BLACK SWAN (DRONAMICS, Bulgaria)	350kg
Fixed Wing	Engine	The NASC Teros (Navmar Applied Science, United States)	272kg
VTOL	Hybrid	Chaparral C1 (ELROY AIR, United States)	250kg
Fixed Wing	Engine	Patroller* (Safran, France)	250kg
Fixed Wing	Engine	U650 (UVSIS, China)	250kg
Advanced Air Mobility	Electric	EH 216 (Ehang, China)	220kg
Helicopter	Engine	BVS VT 500 (Radar MMS, Russia)	150kg
Airship	Electric	Cloudline (Cloudline Africa, South Africa)	100kg
Helicopter	Engine	AWHERO (Leonardo, Italy)	85kg
Fixed Wing	Engine	YZ8 (BMP, China)	80kg

Table 4 shows the world's top 10 in terms of maximum payload. Here, multi-copters are in the top ten along with fixed-wing and VTOL aircraft. In terms of power, electric types are seen in addition to engines and hybrids. Flight speed and flight time are dominated by Europe and the US, but Asian aircrafts are doing well in terms of payload. Three of the top 10 aircraft are Asian: three made in China.



(a) 1st: The NASC Teros (b) 2nd: Patroller



(c) 3rd: Black Swan

Fig. 7 World-class top 3 performance drones

So far, we have looked at the top 10 drones in terms of maximum flight speed, longest flight time, and maximum payload. Ideal flight performance is one that is excellent in all three areas: maximum flight speed, longest flight time, and maximum payload. From this perspective, there are drones that are world-class in all three areas, which are shown in Fig. 7. The American drone "The NASC Teros" is ranked 3rd, 5th, and 2nd in maximum flight speed, longest flight time, and maximum payload, respectively, and is ranked 1st overall. The French drone "Patroller" is ranked outside the top 10, 9th, and 3rd, and is ranked 2nd overall. The third place overall is the Bulgarian drone "Black Swan," which is ranked 4th (200km/h) in maximum flight speed, outside the top 10 in longest flight time, and has the highest maximum payload, and has a range of over 2,500km, so it has a flight time of over 12 hours.

Next, we will look at the flight performance of drones in the world, limited to those that are electrically driven.

Naturally, the performance is lower than that of engine-driven drones. So, when we looked at the top electric flight performance aircraft with a maximum speed and cruising speed of 100km/h or more, a maximum payload of 5kg or more, a flight time of 1 hour or more, and a body weight of 10kg or less, we found that the best electric motor-driven VTOL aircraft, the Wingcopter 198, with a maximum speed of 144km/h, was the one with the highest flight performance, as shown in Table 5.

TABLE 5 Maximum flight speed with electric drive only

Type	Model Name (Manufacturer, Country)	Maximum Speed
 VTOL	Wingcopter 198 (Wingcopter, Germany)	144km/h
 Fixed Wing	Albatros (s Applied Aeronautics, United States)	129km/h
 Helicopter	SARAH 4.0 (Flying-Cam, Belgium)	140km/h
 Multicopter	FALCON (Easy Aerial, United States)	105km/h



(a)Wingcopter (b)Stolkar VXE30

Fig. 8 Maximum speed and Maximum flight time with electric drive only

Fig. 8(a) shows Wingcopter with the Maximum speed 144km/h, the flight time 90 minutes in fixed-wing mode and the maximum payload 5kg. Although the aircraft weights only 10kg, it is relatively light, yet has high speed and endurance. Triple drop system delivers three packages simultaneously, Implements collision avoidance by combining optical recognition, ADS-B, FLARM, remote ID, and AI.

In terms of flight time, excluding wired and solar-powered aircraft, the top electric motor-driven aircraft is the Lockheed Martin Stalker VXE30, which can fly for up to 8 hours and is powered by a solid oxide fuel cell (SOFC). However, since it is still under development, the text is lightened in Table 6. For battery-powered electric motor-driven aircraft, the top three are the Applied Aeronautics Albatross and Alpi Aviation Strix-DF, which can fly for up to 4 hours. Fuel cells are a technology, but they are not yet widespread, and hybrids are being adopted as a transitional technology, especially for VTOL types.

Fig. 8(b) shows Stolkar VXE30 with the maximum speed: 93.3km/h, the flight time 8 hours, the flight range

TABLE 6 Maximum flight time with electric drive only

Type	Model Name (Manufacturer, Country)	Flight Time
 VTOL	Stalker VXE30 (Lockheed Martin, United States) ※Fuel Cells	8 hours
 VTOL	EDGE 130 (FlightWave Aerospace, United States)	2 hours
 Fixed Wing	Albatross (Applied Aeronautics, United States)	4 hours
 Fixed Wing	Strix-DF (Alpi Aviation, Italy)	4 hours
 Helicopter	Velos V3 (Velos Rotors, United States)	1.5 hours
 Multicopter	GeoDrone6 (Nordic Drones, Finland)	1.5 hours

433km. Stolkar uses the propane fuel cell + rechargeable battery allows for up to 8 hours of flight and also equipped with EO/IR camera and low-light camera with target lock function. Used for day and night surveillance and reconnaissance, aerial firefighting support, etc.

However, this is an evaluation of aircraft that have already been developed with a takeoff weight of 10kg or less, but there are some aircraft with improved performance when it comes to aircraft over 10kg. When the takeoff weight is around 10kg to 100kg, there are many aircraft with optimized designs for eVTOL aircraft that approach flight speeds of 150km/h to 250km/h.

In particular, if passenger drone aircraft under development are included, there are many aircraft with flight performance that exceeds that of engine-driven aircraft. For example, the "Vy400R" by Transcend Air of the United States has a design target maximum speed of 660km/h, and the "TriFan 600" by XTI Aircraft of the United States has a design target maximum speed of 555km/h. The "S4" by Joby Aviation, which is said to be the most advanced in development, has a maximum speed of 322km/h. All of these are hybrid or electric based AAM, and most of them are expected to have speeds equal to or faster than engine-driven fixed-wing aircraft.

V. DRONE USE CASES AROUND THE WORLD

Use cases for drones used for civilian purposes around the world are classified as shown in Table 7. A notable feature is that while pesticide spraying is one of the top use cases for drones in Japan, it is not common around the world and, as will be shown later, falls into the precision agriculture category, such as growth surveys, and is classified here as "other".

TABLE 7 Drone usecases

ISR	Intelligence Surveillance and Reconnaissance. Comprehensive on-site information gathering to support decision-making. Disaster relief.
Disaster Relief	Damage assessment, search and rescue for victims, support for first responders (police, fire department, rescue teams), firefighting, forest fire prevention, etc.
Surveillance & Security	Border patrol, coastline monitoring (crackdown on illegal immigrants, etc.), maritime surveillance, poacher surveillance, traffic surveillance, surveillance within facilities to prevent crime, etc.
Survey & Mapping	Site scanning for agriculture, forestry, construction, mining, etc. Data acquisition for creating 2D and 3D maps, vegetation maps, etc.
Investigation & Observation	Environmental monitoring, wildlife surveys, marine surveys, atmospheric observations, etc. (Mapping made for industrial purposes were classified as surveying and mapping.)
Infrastructure Monitoring	Monitoring and anomaly detection for wide-area infrastructure including linear infrastructure such as pipelines, power lines, railways, roads, oil fields, airports and surrounding areas.
Structural Inspection	Close-up inspection of structures, indoors and outdoors, such as steel towers, power lines, power generation facilities, production facilities, bridges, building exteriors and roofs, and the inside of pipes and chimneys.
Delivery	Transportation of goods and medicines, sea and mountain transportation, etc. (When the purpose is limited to rescue, such as dropping life jackets, it is classified as disaster relief.)
Other	Pesticide spraying (rare in Europe and the US), entertainment aerial photography, aerial advertising, anti-drone, bird pest control, pseudo satellites, planetary exploration, etc.

The top use case for VTOL drones was wide-area surveying and mapping at 51%, followed by transporting goods in urban areas in second place and wide-area disaster response, surveillance and security at third place. For fixed-wing drones (large), wide-area surveillance and security was 84%, followed by wide-area investigation and observation in second place and disaster response and infrastructure monitoring in third place. Similarly, for helicopters, the top use case was wide-area disaster response at 72% and security,

etc. For fixed-wing drones (small), the top use case was limited-area surveying and mapping at 74%, while for multicopters, the top use case was limited-area surveillance and security at 43%. As already mentioned, VTOL, fixed-wing drones (large), and helicopters are suited to wide-area flight, while fixed-wing drones (small) and multi-copters are suited to precise flight in limited areas. Looking at each aircraft in more detail, helicopter types, which are also used for transporting goods, are primarily used for non-urban transport such as mountain transport, sea transport, and transporting relief supplies, while many VTOL types are intended for transport between cities and regions. The reason for this is the power source of the propulsion system in Table 1. VTOL aircraft use electric or hybrid propulsion systems and are battery-powered for takeoff and landing. On the other hand, helicopters often use engines as their propulsion system, which means that they are noisy during takeoff and landing, and they are not suitable for transport in urban areas due to exhaust fumes.

Considering the above, and considering the trend toward further strengthening electrification and hybridization against the backdrop of the SDGs, the future of drone logistics is expected to be dominated by eVTOL (electric Vertical Take-Off and Landing) aircraft in inter-city and inter-regional drone logistics that will be realized around the world from 2024 onwards.

Table 8 shows the flight performance of eVTOL aircraft for use cases such as intercity and interregional transportation. All of these are hybrid or electric, and we can see that hybrid types have larger payloads and longer ranges.

TABLE 8 Flight performance of eVTOL for inter-city and inter-regional transportation use cases

Model Name(Manufacturer)	Power Source	Maximum Payload	Range
BELL APT70 (Bell Textron)	Hybrid	45kg	56km
Chaparral (ELROY AIR)	Hybrid	135~250kg	500km
VOLY 20 (Volansi)	Hybrid	13.5kg	Over 563.3km
Aero2 (Dufour Aerospace)	Hybrid	20~0kg	400km
Nuova V300 (PIPISTREL)	Hybrid	460kg	300km
Wing (Wing)	Electric	1.2kg	20km
Manta Ray (Phoenix-wings)	Electric	7~10kg	80~120km
Wingcopter 198 (Wingcopter)	Electric	5kg	75~10km
Avy Aera (Avy)	Electric	3kg	80~100km
DeltaQuad Pro (Vertical Technologies)	Electric	1.2kg	120km
Fixer 007 (FIXAR Global)	Electric	2kg	60km

VI. ADVANCED AIR MOBILITY (AAM)

The concept of a AAM also known as an eVTOL, a vertical take-off and landing aircraft, has been around since 2010, but a major turning point was the concept of "Uber Air" proposed by Uber Technologies in the fall of 2016. The target settings are a speed of 240-320 km/h, a range of 60 miles (96 km) on a single charge, and the number of passengers is becoming the same as that of a car, as shown in Fig. 9. This is the field where the most active evolution is taking place among drones,

In North America, the Federal Aviation Administration (FAA) and in Europe, the European Union Aviation Safety Agency (EASA) are driving the evolution of flying cars. When taking a general look at the current state of AAM in Europe and the US, we notice that many of them have a

configuration similar to fixed-wing aircraft. If we roughly divide these into those with or without fixed wings, as shown in Fig. 9, 60% have fixed wings and 40% do not, so winged aircraft are dominant. These analyses were based on a database of 41 manufacturers of AAM in the world.

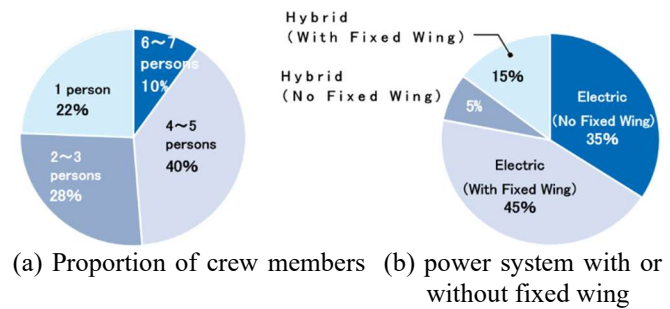


Fig. 9 Passenger capacity and power source system share with passenger drone type (with or without fixed wing)

The goal setting of the "Uber Air" project of Uber Technologies, which has made a great contribution to the development of eVTOL in the United States, is assumed to be the de facto standard. Many AAM similar to fixed-wing aircraft, including Joby Aviation's "S4", are expected to meet this goal. Another reason seems to be the result of the discovery of the value of AAM as a means of transportation to places that are too far to reach by car but too close to reach by passenger plane. NASA's "Advanced Air Mobility Project" aims to provide innovative aviation systems to "places where aviation has not been used or where services are inadequate", and supports the development and social implementation of eVTOL aircraft. On the other hand, Volocopter's "VoloCity", which is considered to be one of the leading companies in the development of AAM, emphasizes safety, reliability of control, and economic efficiency as a means of transportation in urban areas that replaces cars and taxis, transporting passengers to major hubs such as train stations and airports, and has solid performance of two passengers, a maximum speed of 110 km/h, and a flight distance of 35 km. In addition, many of the ultralight aircraft that are limited to sports and leisure use are also multi-copters. The difference in characteristics between the two types is expected to become clearer as AAM are implemented in society.

Table 9 shows the top AAM aircraft (maximum speed, all under development), and Fig. 10 shows the first and second place aircraft, which are still under development. Both are American-made, have a speed of over 500 km/h, a flight time of over 1 hour, and a crew of about 6 people. Therefore, the flight range is over 500 km to 600 km, which is almost the same as the specifications of a small private jet aircraft. Table 10 shows the top 10 AAM in terms of flight time. A Chinese manufacturer is in 8th place.

The flying car field is the subject of the most fierce development competition within the drone industry, and as of May 2023, there are 51 flying car manufacturers in the world (41 have conducted aircraft surveys). The United States has the most with 17, followed by the UK, Germany, and China with four, Japan and Israel with three, France and Australia with two, and Sweden, the Czech Republic, South Korea, India, Poland, Hungary, and Liechtenstein with one each.

TABLE 9 Top 10 AAM flight performance (Top speed, all under development)

Power Source	Fixed Wing	Model Name (Manufacturer, Country)	Maximum Speed
Hybrid	Yes	Vy400R (Transcend Air, United States)	660km/h
Hybrid	Yes	TriFan 600 (XTI Aircraft, United States)	555km/h
Electric	Yes	MOB-One (Airspace Experience Technologies, United States)	482km/h
Hybrid	Yes	e-STARLING (SAMAD Aerospace, United Kingdom)	463km/h
Electric	Yes	VA-X4 (Vertical Aerospace, United Kingdom)	325km/h
Electric	Yes	S4 (Joby Aviation, United States)	321km/h
Hybrid	Yes	J-2000 (Jetoptera, United States)	321km/h
Hybrid	Yes	Zurij (Zurij, Czech Republic)	320km/h
Electric	Yes	Orca eVTOL (Orca Aerospace, Hungary)	300km/h
Electric	Yes	VERTIIA (AMSL Aero, Australia)	300km/h

TABLE 10 Top 10 AAM flight performance (Maximum flight time, all under development)

Power Source	Fixed Wing	Model Name (Manufacturer, Country)	Flight Time
Electric	No	Skai (Alakaï Technologies, United States)	4 hours
Hybrid	Yes	VERTIIA (AMSL Aero, Australia)	3 hours
Electric	Yes	Atea (Ascendence Flight Technologies, France)	2.5 hours
Hybrid	Yes	TriFan 600 (XTI Aircraft, United States)	2 hours
Hybrid	Yes	e-STARLING (SAMAD Aerospace, United Kingdom)	2 hours
Hybrid	Yes	Zurij (Zurij, Czech Republic)	2 hours
Hybrid	Yes	Vy400R (Transcend Air, United States)	1.5 hours
Electric	Yes	PROSPERITY I (Auto Flight, China)	1.5 hours
Electric	Yes	ASKA (ASKA, Israel)	1.5 hours
Hybrid	Yes	J-2000 (Jetoptera, United States)	1 hours



(a) Vy400R (b) TriFan600
Fig. 10 Top level passenger drone

Of these, 18 manufacturers are conducting test flights at the actual aircraft level with people on board, while the remaining 24 companies are only at the level of flight tests using scale models. The 18 companies with the highest level of completion are 9 from the United States, 3 from China, 2 from Germany, 1 from the United Kingdom, France, Sweden, and the Czech Republic. All 18 companies are electric or hybrid, with 62% being VTOL-type winged aircraft and 38% being wingless, showing the superiority of winged aircraft. In addition, according to an explanation given to shareholders on August 3, 2023, the "VA-X4 (commonly known as VX4)" in Table 9, which is being developed by Vertical Aerospace in the UK, crashed during flight tests. Fortunately, there were no casualties and only some damage was caused to the aircraft. This flight test was conducted under the assumption of a motor failure, and it appears that the crash occurred from an altitude of 6m. According to the explanation given to shareholders, it seems that manned flight tests were scheduled to be conducted within the next few months, but flight tests have been suspended until the cause of the accident is identified and measures are taken. An application for type certification has already been submitted to the Japanese Ministry of Land, Infrastructure, Transport and Tourism's Civil Aviation Bureau, and it is known that Japan Airlines and Marubeni have placed pre-orders for the VX4. Well, due to the inability to obtain permission from the Civil

Aviation Bureau, only demonstration flights without passengers are planned for the 2025 Osaka Kansai Expo.

As mentioned above, the catalyst for the global AAM industry was the aforementioned report published by the American company Uber in 2016. The report shocked the world by calculating that in the future, AAM could be used at a cost lower than that of a private car. The world's most promising aircraft development venture is Joby Aviation. Joby Aviation was founded in California in 2009, acquired the AAM business from Uber in 2020, and listed on the New York Stock Exchange as a SPAC in 2021. It plans to develop a five-seater eVTOL aircraft and commercialize it in 2024. Toyota Motor Corporation has a 13% investment.

Fig. 11 shows the global AAM business market and growth forecast. This is forecast data from Morgan Stanley (April 2021), which predicts that the market will grow to a huge size of approximately 1.5 trillion US dollars by 2040. At the 2023 level, the market size will be approximately 24.7 billion US dollars, but since services are essentially close to zero, a significant revision will be necessary.

Yano Research Institute, which published its survey results in 2023, has made a realistic prediction, estimating that the global market will be about 0.4 billion US dollars in 2025, about 6.7 billion US dollars in 2030, 255 billion US dollars in 2040, and 1.2 trillion US dollars around 2050. Morgan Stanley's forecast for 2040 is 1.5 trillion US dollars, so Yano Research Institute's growth forecast for 2040 is about 1/6 of that. Recalling that the growth forecast for drones in 2030 shown in Fig.1 was 55.8 billion US dollars, it is expected that the market size will be comparable to that of the drone industry. No one can imagine at this point that a market of 6.7 billion US dollars will be generated from almost nothing in seven years, but revolutionary innovation is happening all over the world right now.

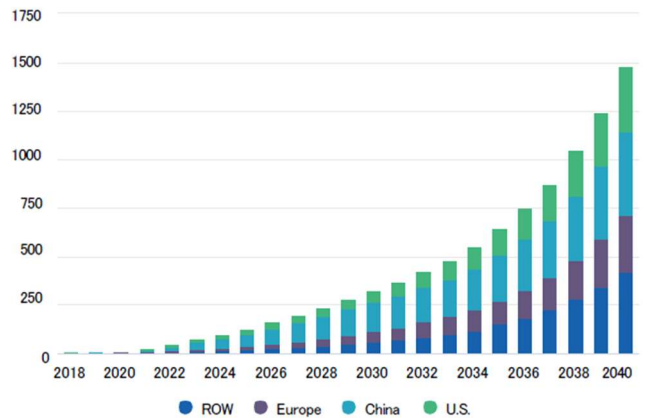


Fig. 11 AAM global total forecast addressable market, provided by Morgan Stanley Research Estimate

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